Name: ______________________

Worksheet

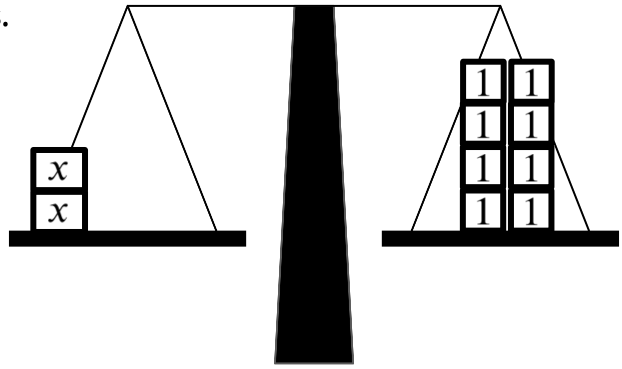
Rearranging to solve linear equations

Task A: Representing equations with unknowns on both sides

1) Build an equation with unknowns on both sides.

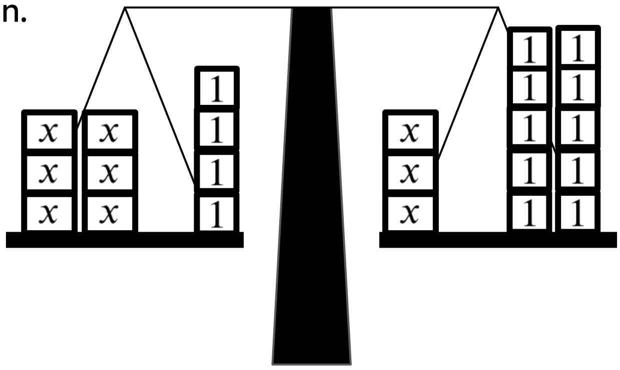
Starting with this balance showing $2x = 8$ show what you need to add to both sides to make the equation $7x + 2 = 5x + 10$

$$2x \quad = \quad 8$$



2) ‘Un-do’ this balance model to solve this equation.

$$6x + 4 \quad = \quad 3x + 10$$



3) Use this bar model to solve this equation.

Write your algebraic notation alongside.

$$2x + 161 \quad = \quad 8x + 29$$

<table>
<tr><td>x</td><td>x</td><td colspan="7">161</td></tr>
<tr><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>29</td></tr>
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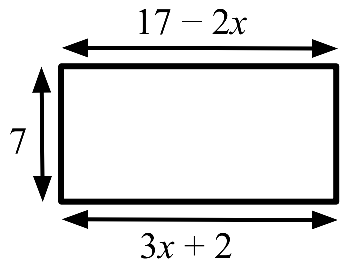
Task B: Solving equations with unknowns on both sides

1) Solve this equation in two different ways:

a) $3x + 2 = 7x - 10$ $\qquad$ $3x + 2 = 7x - 10$

b) $2x + 2 = 23 - 5x$ $\qquad$ $2x + 2 = 23 - 5x$

2) Find the perimeter of this rectangle.



3) Solve: $\frac{1}{8}x - 7 = 3 - 7x$

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